INCH-POUND
MIL-S-63433D (AR)
AMENDMENT 3
14 March 1996
SUPERSEDING
AMENDMENT 2
10 April 1990

MILITARY SPECIFICATION

SIMULATOR, ARTILLERY FLASH, M21
PERFORMANCE SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-S-63433D (AR), dated 20 January 1989, and is approved for use by the U.S. Army Armament, Research, Development and Engineering Center and is available for use by all Department and Agencies of the Department of Defense.

PAGE 2

2.11: Add to list of Military Standards:

"MIL-STD-1234 Pyrotechnics, Sampling, Inspection and Testing".

PAGE 3

* 3.1 Assembly, Line 3: Delete "Simulator Tank Gunfire Device 17-61, EC Corporation device designated as Simulator, Tank, Gunfire Device 17-157, and the Gunfire Simulator, Device 17-133 manufactured by Joanell Laboratories," and substitute "Simulator, Tank, Gunfire Device 17-61 and EC Corporation device designated as Simulator, Tank, Gunfire Device 17-157,"

PAGE 4

3.2.4.5, Electrostatic discharge: Change to "Electrostatic discharge - composition".

PAGE 5

3.2.5.1: Revise as follows:

DOD Hazard Classification. - Delete "1.3" and substitute "1.4".

DOT Explosive Label - Delete "Class B" and substitute "Class C".

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DOT Marking - Delete "SPECIAL FIREWORKS" and substitute "SIGNAL FLARES".

PAGE 6

3.3.6:

Line 3: Delete "applicable gunfire devices" and substitute "socket contacts of the connector block (see 6.5.3)."

* Connecting Wire Assembly Resistance: Delete "1.51 ohms max." and substitute "1.1 ohms to 1.6 ohms".

PAGE 7

* 3.4.1.1 Sound characteristics: Delete "96 meters min" and substitute "96 ± .5 meters"

PAGE 9

3.5.2: Five foot drop: Delete "4.5.8.12" and substitute "4.5.8.4".

PAGE 10

3.5.12: Delete in its entirety and substitute the following:

"3.5.12 Heat soak at 160°F. (First Article Test Only) The packaged simulators (in barrier bag) shall meet the functioning requirements of 3.4.1 after being subjected to a temperature of 71.1 ± 3°C, (160°F + 5°F) for a minimum period of 5 weeks. Prior to functioning, the simulator assembly resistance shall not exceed 2.0 ohms maximum."

3.5.13: Delete in its entirety and substitute the following:

"3.5.13 Temperature-humidity. (First Article Test Only). The packaged simulators (in barrier bag) shall meet the functioning requirements of 3.4.1 after exposure to temperature-humidity cycling as specified in 4.5.8.10. Prior to functioning, the simulator assembly resistance shall not exceed 2.0 ohms maximum."

PAGE 12

* 4.3.3.b: Delete "(12 functioning tests for a total of 918 samples)" and substitue "(7 functioning tests for a total of 792 samples)." Delete "918-30-31" and substitute "792-26-27"

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PAGE 13

Table I: Add under "Paragraph ref./Insp. method" column:

"4.4.3.1" to "Charge composition" and "4.4.3.1" to "Paper or paper products". Delete "4.5.4.4" and "4.5.4.5"; and substitute "4.5.14" and "4.5.15".

PAGE 14

Table I:

Temperature - humidity: Delete "3.5.12" and "4.5.8.9" and substitute "3.5.13" and "4.5.8.10".

Temperature cycling: Delete "Temperature cycling, 3.5.13, 4.5.8.10/4.5.4" and substitute "Heat soak at 160°F, 3.5.12, 4.5.8.9/4.5.4".

* Under Non-Sequential Test, delete "Temperature Cycling" in its entirety and substitute "Heat Soak at 160°F , 54, 1-2, 3.5.12, 4.5.8.9/4.5.4".

PAGE 18

4.4.2.1: Add under "Paragraph ref/Insp method" column:

"4.4.3.1" to "Charge Composition" and "4.4.3.1" to "Paper products".

PAGE 20

4.5.2, Line 2: Delete "approved gunfire device" and substitute "firing tubes of a Hoffmann Device".

PAGE 21

4.5.4, Functioning: Line one, change "6" to "12".

4.5.4, Line 4: Delete "an approved gunfire device" and substitute "a Hoffmann Device".

4.5.4.1, Line 3: Delete "suspended" and substitute "fired using a Hoffmann Device, mounted horizontally,"

PAGE 22

4.5.8.2, Cold temperature: Line three, change "6" to read "12".

4.5.8.3, Hot temperature: Line three, change "6" to read "12".

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PAGE 23

4.5.8.9: Delete in its entirety and substitute the following:

"4.5.8.9 Heat soak at 160°F. (First Article Test Only) The packaged simulator assemblies shall be conditioned for a period of five(5) weeks minimum at the specified temperature. After removal from the conditioning chamber and allowed to return to ambient temperature, resistance measurement shall be made using a Fluke 8024B, 8050A Multimeter, or equivalent meter, approved for use with electro-explosive devices.

SAFETY NOTE: All resistance measurements shall be conducted from behind a barricade and shorting caps shall be in place at all times except during resistance measurement and when functioning items.

The simulator resistance shall not exceed the applicable requirement after conditioning. Only those simulators passing the resistance requirement shall be placed in the firing device and functioned (ref. 4.5.4). Any simulator failing to meet the applicable requirements shall be classed defective (Destructive test)".

4.5.8.10 Delete in its entirety and substitute the following:

"4.5.8.10 Temperature humidity. (First Article Test Only) The packaged simulators shall be subjected to a 28-day temperature-humidity test in accordance with MIL-STD-331, Test 105.1, para. 5.2, Single Chamber Cycling Method. After removal from the conditioning chamber and allowed to return to ambient temperature, resistance measurements shall be made and the simulators functioned as specified in 4.5.8.9 above. Any simulators failing to meet the applicable requirements shall be classed defective. (Destructive test)."

PAGE 24

* 4.5.8.11 Water immersion test: delete "(Non-destructive test)".

PAGE 27

4.5.15: Change to "4.5.16"

Add new paragraph:

"4.5.15 Electrostatic discharge. The simulator, bare, packaged and in the Hoffmann Device shall be tested in accordance with Test 126 of MIL-STD-331. Observations shall be made for compliance with the applicable requirements. (Non-

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destructive test)"

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6.4: Under "Paragraph no." column: Delete "4.5.15" and substitute "4.5.16". Delete "6.2.2" and substitute "6.2.1". Add "6.3" in "DI-R-1714" line.

PAGE 29

6.5.3: Delete in its entirety and substitute:

"6.5.3 Interface devices: Hoffmann Device (simulator, Tank Gunfire Device 17-61) for use in chamber withdrawal, functioning, electromagnetic radiation and electrostatic discharge tests. Request from Contracting Officer, Armament Munitions and Chemical Command, Rock Island, IL 61299.

Connector Block with Socket Contacts - 8602D03016 (part of EC Corp Device 17-157) for assurance that simulator electric plug fits all firing devices power outlets. Request from ARDEC, ATTN: AMSTA-AEE-P, Picatinny Arsenal, NJ 07806-5000. (This part is manufactured by the EC Corporation, 10511 Hardin Valley Road, Knoxville, TN 37932)".

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